means every engine you fly is a new engine as what you have tested is but a similar engine, not the same. Liquid engines can control the flight of the rocket through gimbaling; a solid engine needs external control systems which incur more cost and additional weight. If India realised these benefits, perhaps, its PSLV would have been an all-liquid and more versatile rocket. One need not look farther than our GSLVs, which have all its engines propelled by liquid fuel—but for the solid strap-on for initial thrust—to understand this.

Solid gives good initial thrust for lift-off, but it burns out much earlier than a liquid engine. A crucial engine was far superior to both, in terms of burning duration and energy quotient. A liquid engine can be stopped and restarted during tests and flights, ensuring better fuel efficiency; a solid engine once ignited, cannot be stopped. Liquid propellant engines can be tweaked and upgraded for different uses; a solid is virtually as rigid as the fuel.

Propellants are not just to take a satellite to space; they are needed to keep it in orbit. Once in a designated orbit, the satellite moves around the earth by forces governed by gravitational pulls of the earth and other celestial objects, but it tends to stray from this path. Liquid propellants come in handy here, as they can be fired for very short durations and stopped at will. Position sensors and onboard computers work in tandem to fire small control rockets fitted to the satellite to push it back to the orbit whenever it tends to deviate. Geosynchronous satellites that orbit the earth at an altitude of 36,000 km, often swing wildly on different planes. To keep such swings below 3 km from the orbital plane, several control rockets fitted on different planes of the satellite are fired.

The French considered all these—and more—while configuring LIIIS. It was decided that the rocket would have a one-tonne payload. In those days, calculations of the rest of the rocket were made on scientific gut feeling; today you have mechanisms for precise calculations of every stage and component. So, it was presumed that a cluster of four M-40 engines could form the first stage booster. The second stage was to be a single M-40